

Tracy Duchia Yang

(651)-366-9307 | tdyang08@gmail.com | www.linkedin.com/in/tracy-yang-robotics

PROFESSIONAL SUMMARY

Robotics Engineer with 1 year of industry experience in medical robotics and embedded systems. Specialized in ROS/ROS2 development, multi-agent coordination, SLAM, and motion planning. Proven track record of reducing system failures by 25% through novel communication protocols and acceleration development cycles by 52%. Skilled in Python, C/C++, real-time systems, and hardware-software integration.

TECHNICAL SKILLS

Robotics & Perception: ROS/ROS2, SLAM (A*, AMCL, Frontier-Based Exploration), Motion Planning, Path Planning, LiDAR, Computer Vision, OpenCV, Sensor Fusion, TurtleBot3, Multi-Agent Systems, Swarm Robotics

Programming: Python, C/C++, MATLAB, Verilog, Embedded C

Controls & Kinematics: PID Control, Forward/Inverse Kinematics, Velocity Kinematics, AMCL Localization, State Machines

Embedded & Hardware: RTOS, I2C, SPI, CAN Bus, UART, FPGA (Vivado), PetaLinux, Raspberry Pi, Arduino

Tools & Platforms: Git/GitHub, Ubuntu/Linux, Solidworks, OnShape, Gazebo, OpenMV, VSCode

WORK EXPERIENCE

Embedded Design Engineer, Meniscense (Medical Robotics Startup), Hudson, MA May 2024 – Nov 2025

- Architected and automated FPGA build pipelines (Linux, Python, SQL), cutting development cycles by 52% and accelerating research and development
- Programmed control boards and motor controllers (Python, Vivado, PetaLinux), enabling reliable I2C/SPI communication for robotic actuator coordination
- Re-engineered I2C memory mapping architecture, improving embedded system reliability and reducing integration debugging time
- Implemented CAN bus multi-master control for multi-axis robot coordination, reducing acquisition times by 27%

Teaching Assistant, Electrical and Computer Engineering Department, WPI Mar. 2024 – May 2024

- Instructed 40+ students in digital circuit design, state machine implementation, and Verilog HDL programming
- Supervised hands-on labs for FPGA-based control system testing on TI development boards

PROJECTS

Symbiotic Multi-Agent Construction (SMAC), Major Qualifying Project (MQP), WPI Aug. 2024 – May 2025

Technology Used: Python, ROS, Ursina, Solidworks

- Led 8-month development of decentralized robotic construction system using 5-DOF “inchworm” robots and embedded “smart blocks” for scalable, fault-tolerant multi-agent construction
- Architected novel real-time ROS communication protocols, mitigated system failure by 25% through root cause analysis of 3 communication breakdowns
- Developed 3D simulation supporting 10+ cooperative agents with real-time swarm coordination and adaptive path planning

Autonomous Navigation Robot, Unified Robotics IV, WPI Oct. 2023 – Dec. 2023

Technology Used: ROS, Python, TurtleBot3, LiDAR

- Implemented SLAM algorithms (A*, AMCL, frontier-based exploration) to solve “kidnapped robot problem” on TurtleBot3, completing exploration in 8 minutes, 20% faster than class average
- Optimized PID control systems for improved localization and dynamic obstacle handling (ROS, Python)

Pick and Place Robotic Arm, Unified Robotics III, WPI Aug. 2023 – Oct. 2023

Technology Used: MATLAB, OpenMV

- Derived Denavit-Hartenberg parameters and Jacobian matrix for 4-DOF robotic arm, implementing numerical inverse kinematics solver for real-time trajectory planning and enabling coordinated multi-joint manipulation
- Developed computer vision pipeline achieving 98% pick-and-place success rate with live object tracking, color identification across 4 categories, and non-ball object rejection

EDUCATION

Worcester Polytechnic Institute (WPI), Worcester, MA

- Bachelor of Science in Robotics Engineering | GPA 3.50/4.00 May 2025
- Master of Science in Robotics Engineering Expected Graduation: May 2027

Relevant Courses: Robot Dynamics, Motion Planning, AI in Robotics, Industrial Robotics, Robot Controls, Advanced Digital System Design, Real-Time Embedded Systems

CERTIFICATIONS

- **Modern Robotics Course 1: Foundations of Robot Motion, Northwestern University (Coursera)** March 2026